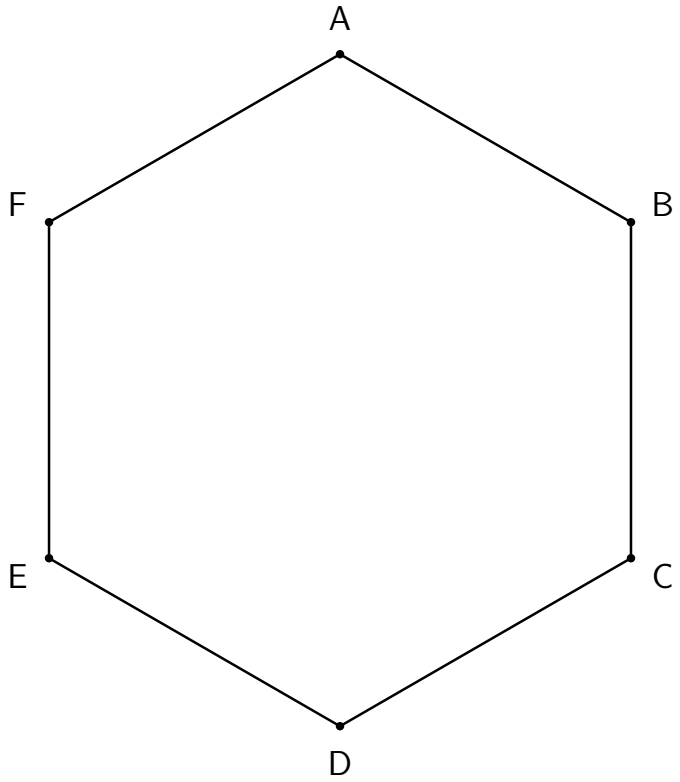
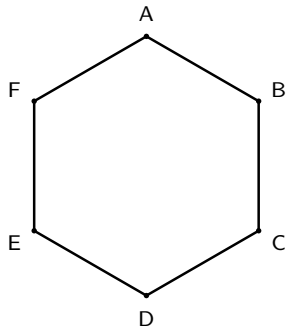
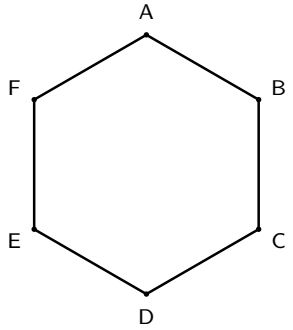
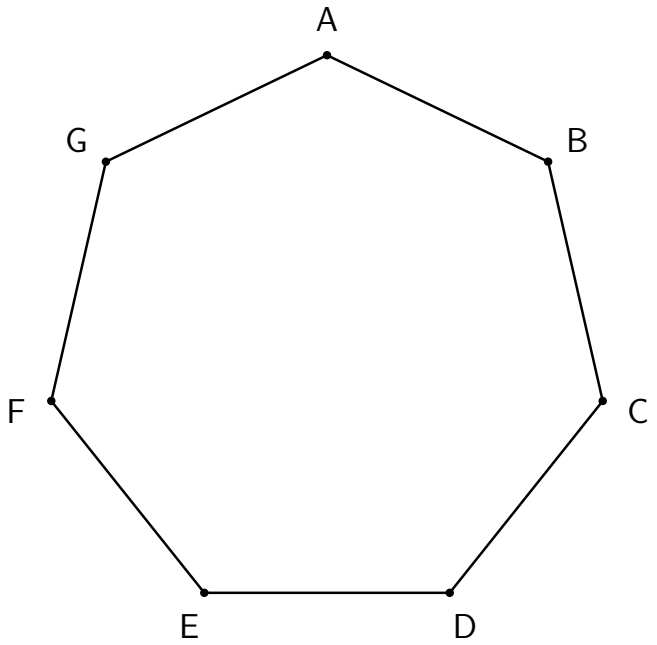
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

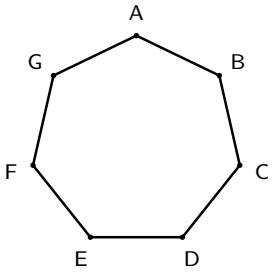
**Problem 1:** Fill each polygon in two different ways. Can the two fillings of one polygon have different numbers of triangles?

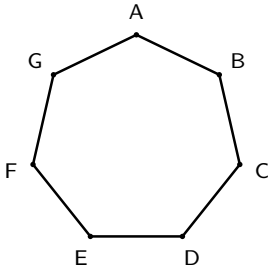


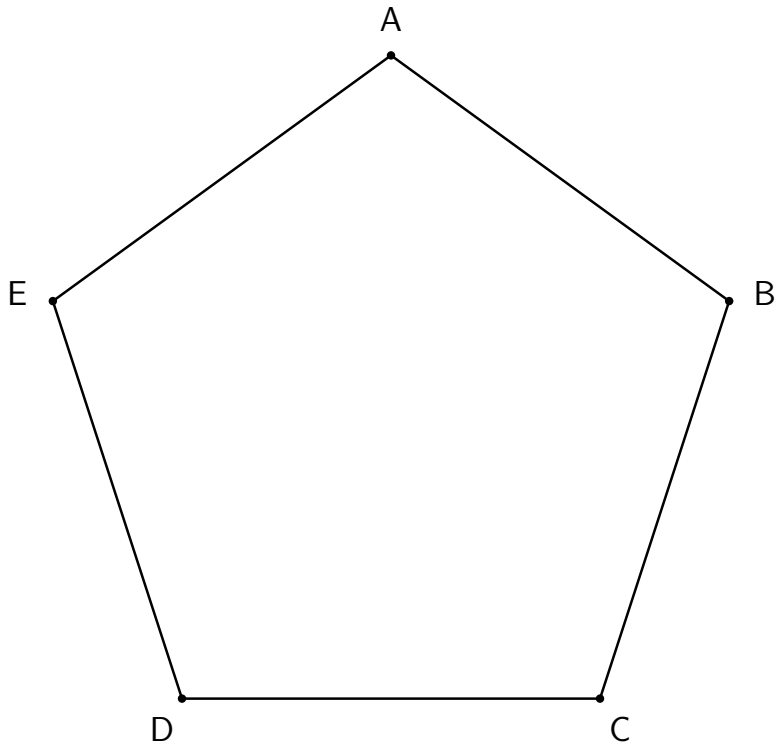



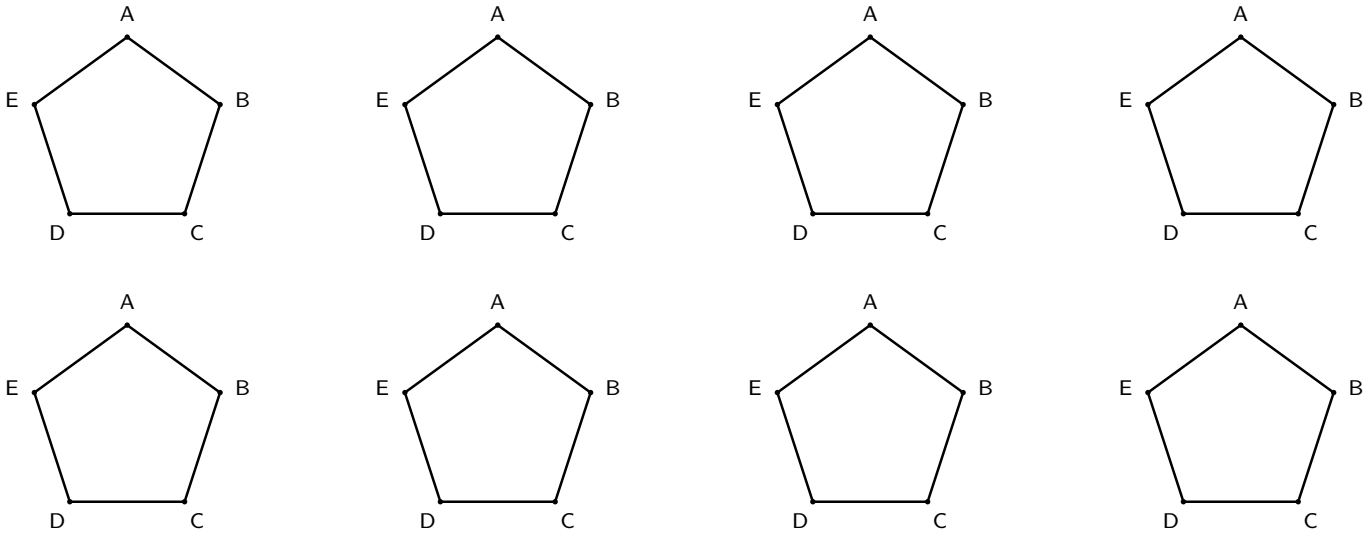



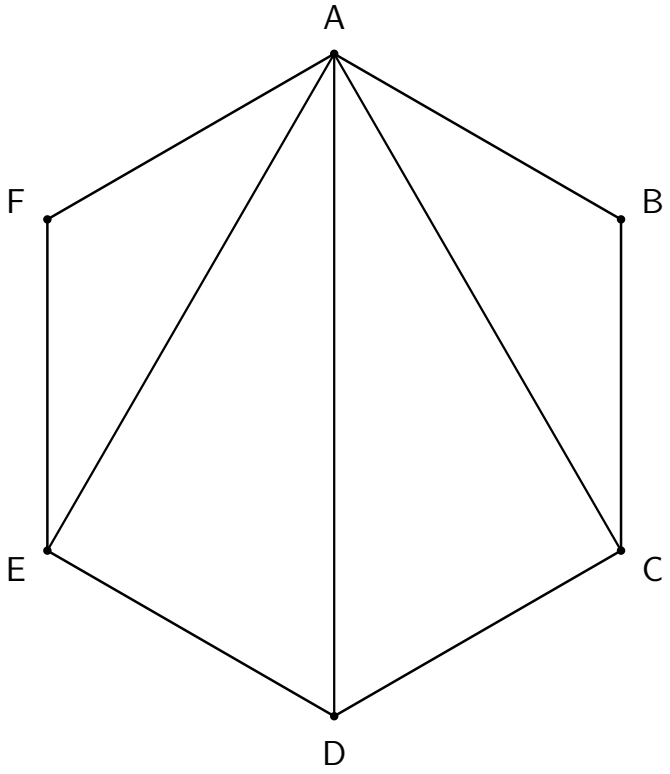



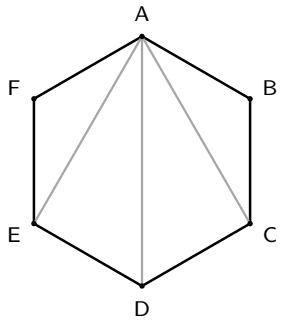
**Problem 2:** Find every filling of this five-corner polygon. How do you know none are missing?

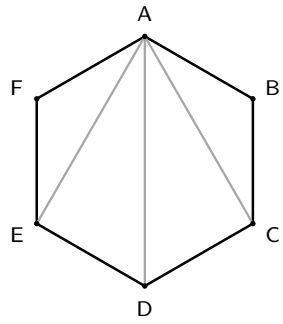


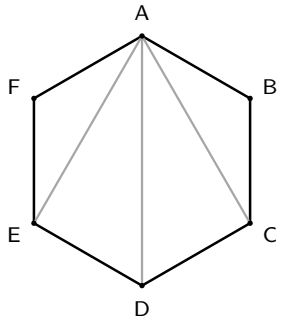



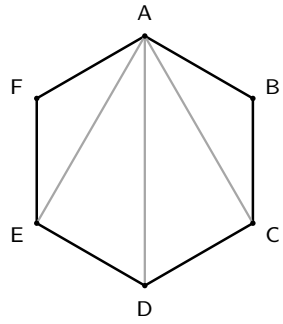
**Problem 3:** Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.

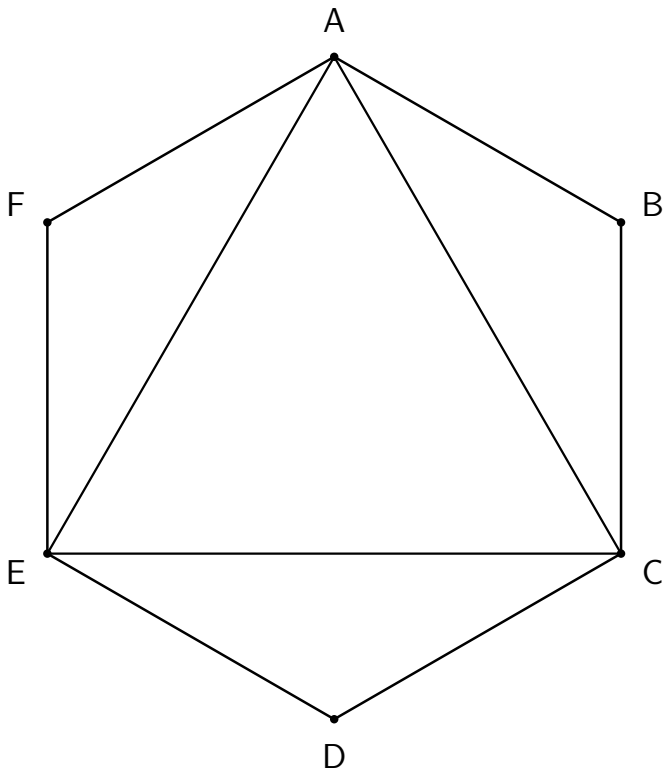


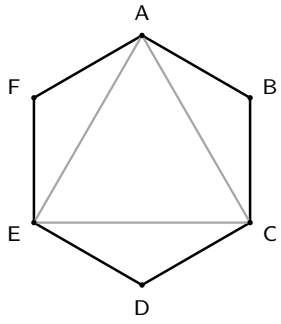



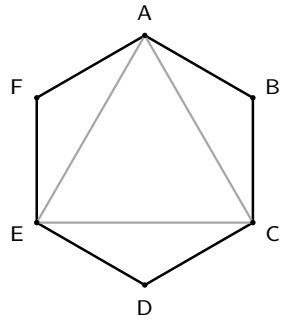



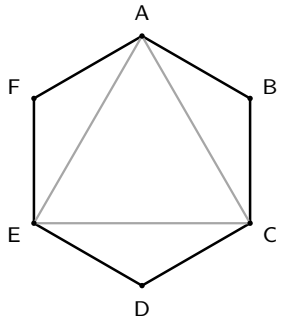



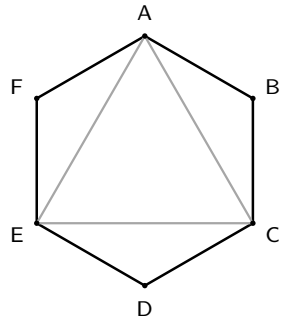



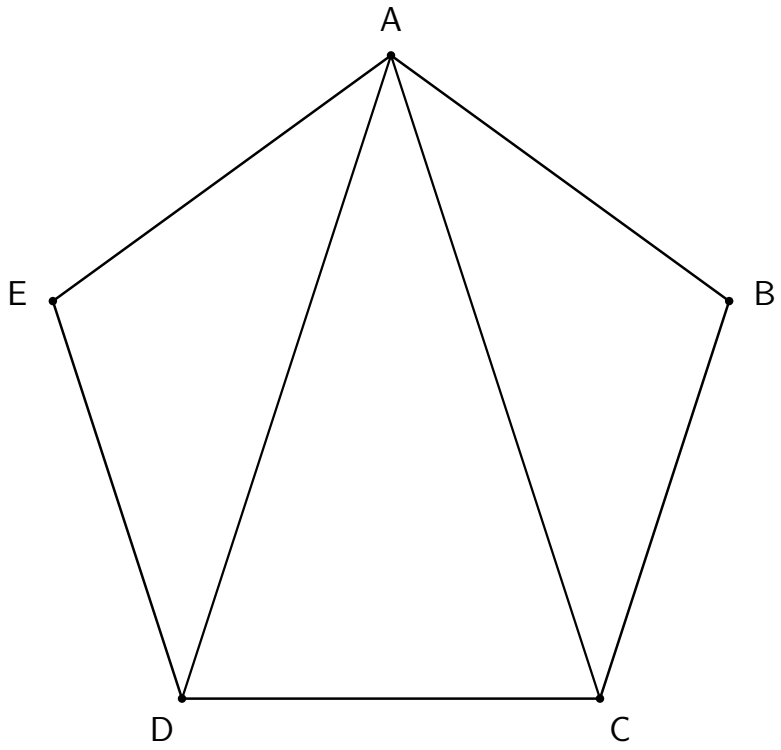



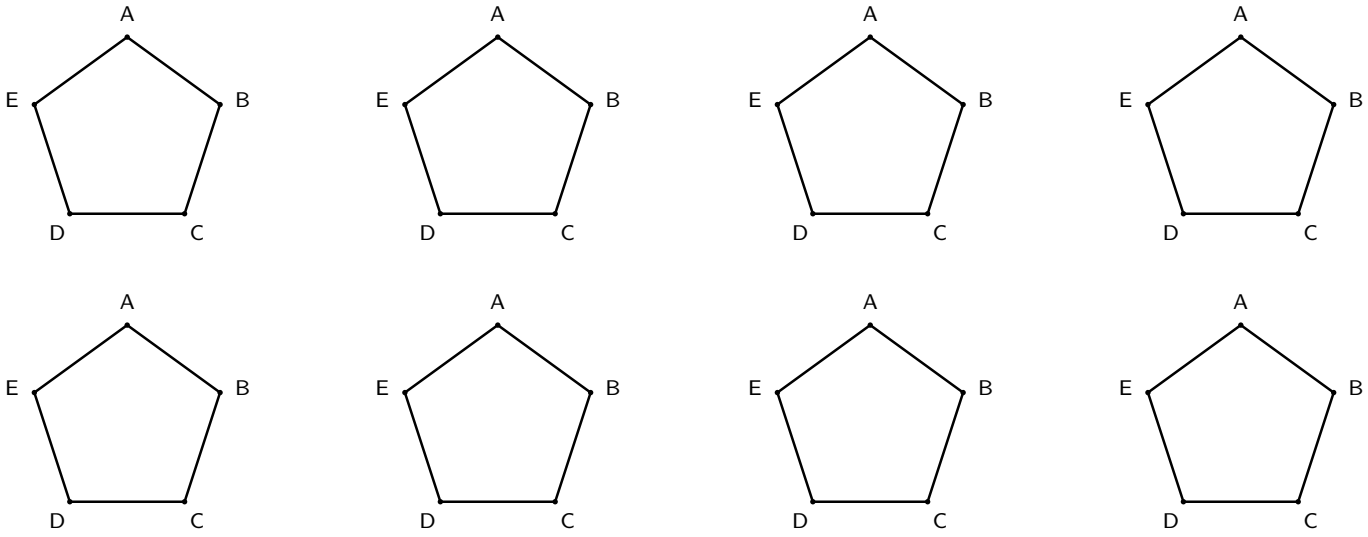



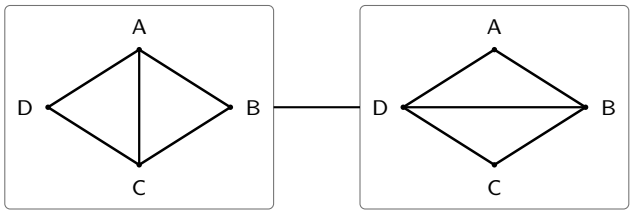



**Problem 4:** Start with this filling. Change one inside line at a time, keeping the polygon filled with triangles, and get back to this filling after an odd number of changes. Show every filling along your route.

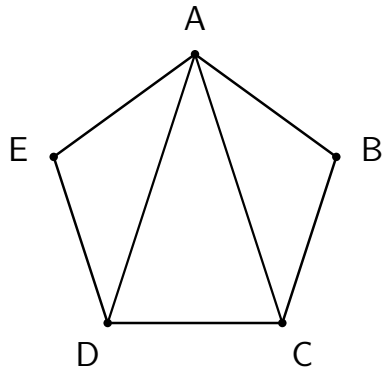


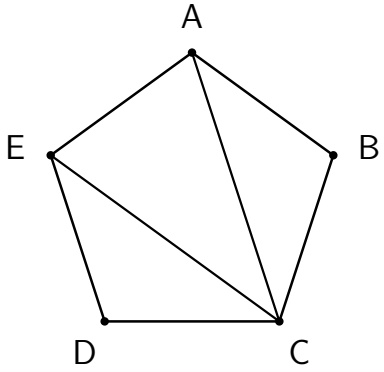



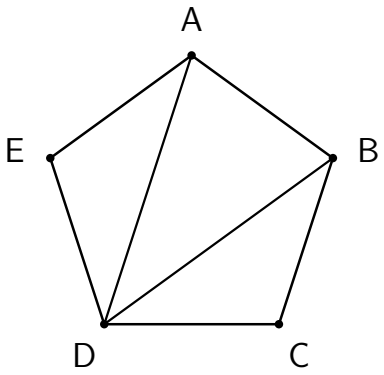
**Problem 5:** A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join.

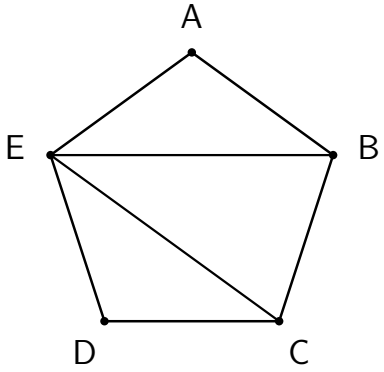


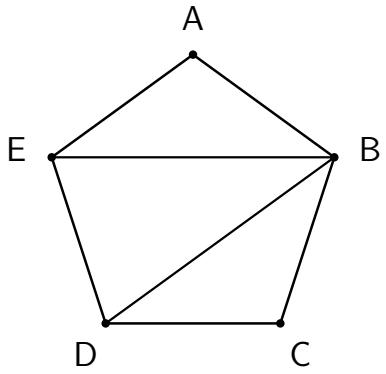

Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.



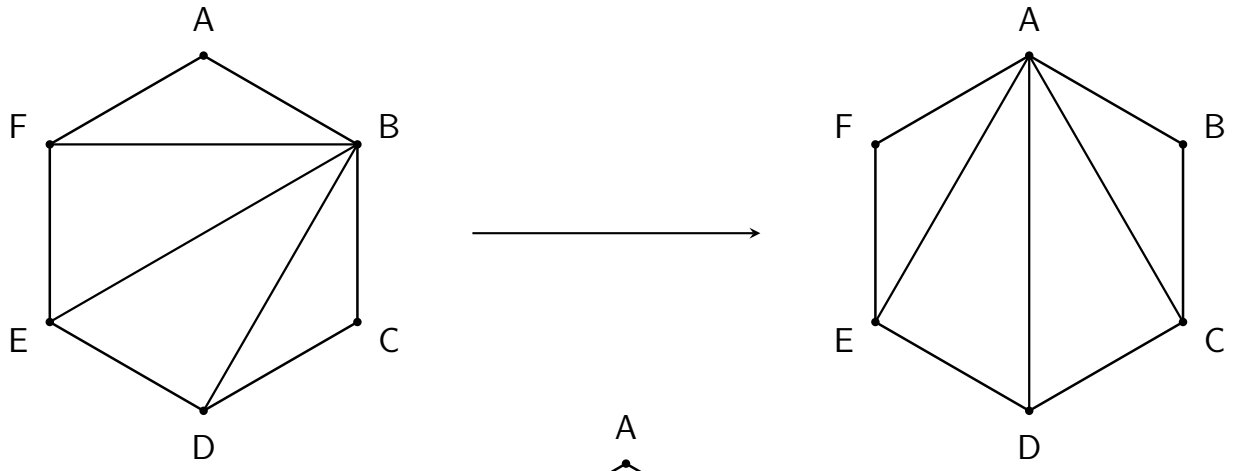



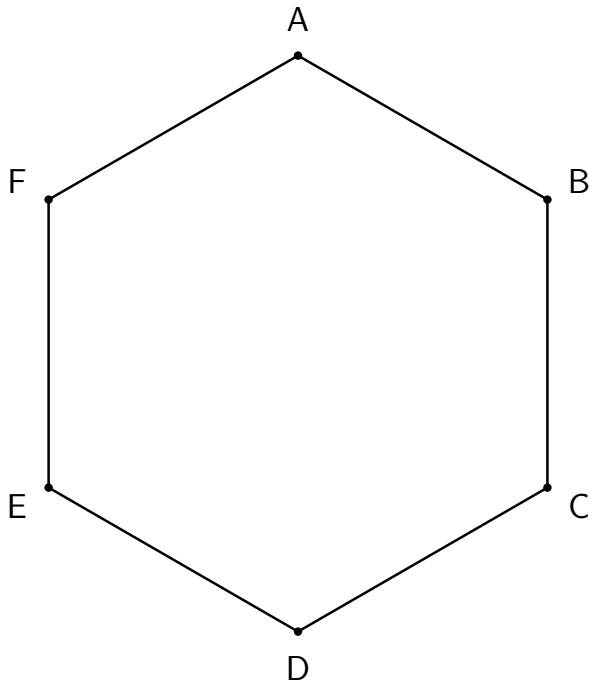



**Problem 6:** Change the first filling into the second using as few flips as you can. Find another route with a different number of flips, without visiting any filling twice.

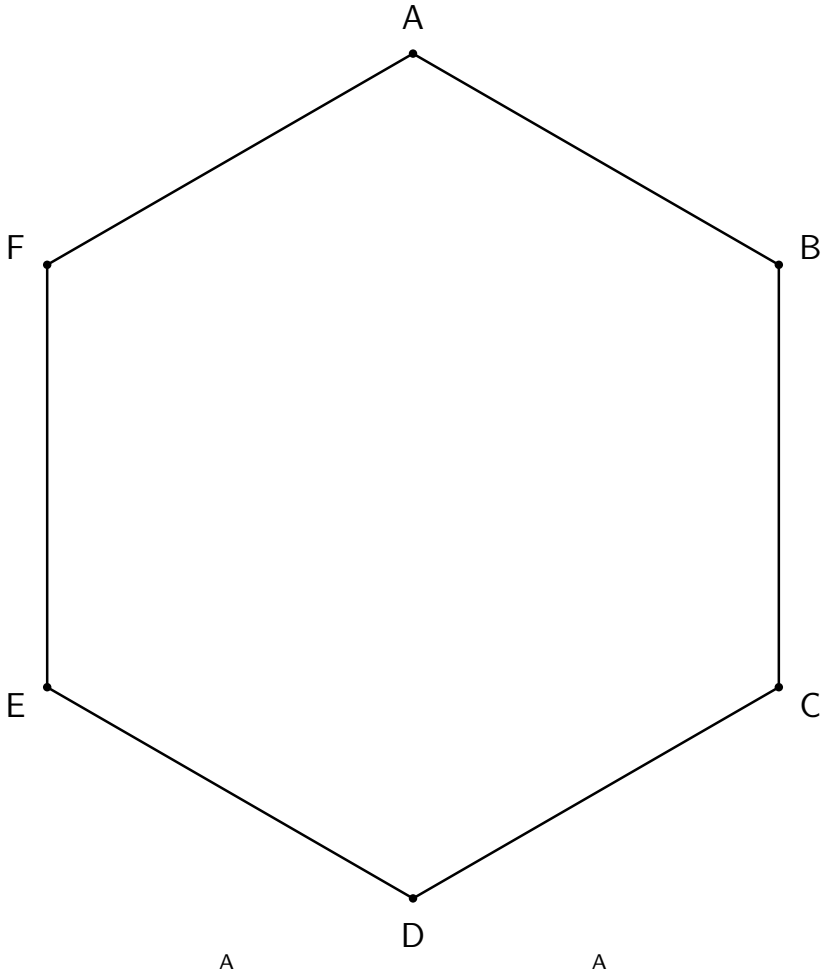


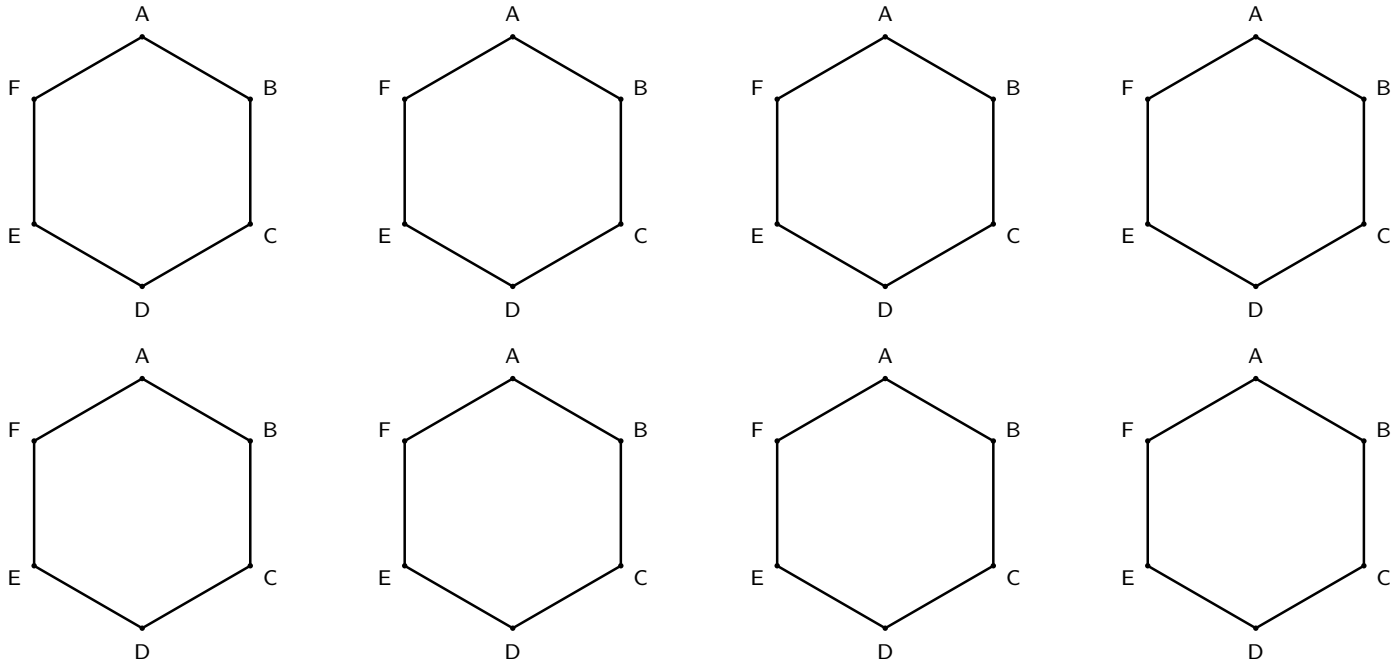




Fewest flips

Another route

**Problem 7:** Find every filling of this six-corner polygon. How do you know none are missing or repeated?



## Problem 7 (continued)

